

AI/ML - Course outline (6th Semester)

Course Code: CS3233

Name of the Programme: B.Tech(Hons.)

Name of the Course: Fundamentals of Natural Language Processing

Semester: 6

Course credit	No. of hours per week	Total no. of Teaching
	(L + T + P)	hours
3	2+0+2	60

Course Objectives:

Introduce students to the foundational concepts of linguistic structures and classical Natural Language Processing (NLP) techniques.

Develop students' ability to design and train deep learning models for solving real-world language processing problems.

Empower students with advanced skills in leveraging transformer-based architectures and transfer learning for cutting-edge NLP tasks.

Provide experiential learning opportunities through practical implementation of speech-enabled, multilingual NLP systems using state-of-the-art open-source tools.

Module – 1 Foundations of NLP & Embeddings No. of hours 6

Text Preprocessing: Tokenization, Lemmatization, Part-of-Speech (POS) Tagging, Named Entity Recognition (NER).

Language Modelling: N-Grams, Perplexity.

Traditional Representations: Bag-of-Words (BoW), Term Frequency-Inverse Document Frequency (TF-IDF).

Modern Embeddings: Word2Vec, Global Vectors for Word Representation (GloVe), FastText.

Contextual Embeddings: Embeddings from Language Models (ELMo), Bidirectional Encoder Representations from Transformers (BERT) embeddings.

Module – 2 Deep Learning for NLP No. of hours 6

Convolutional Neural Networks (CNNs) for Text.

Sequence Models: Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Attention Mechanism.

Evaluation Metrics: Bilingual Evaluation Understudy (BLEU), Recall-Oriented Understudy for Gisting Evaluation (ROUGE), F1, Precision, Recall.

Use Case: Text classification (news, emails, etc.).

Module – 3 Transformers & Foundation Models No. of hours 6

Transformer Architecture.
Pretrained Models: BERT, Robustly Optimized BERT Pretraining Approach (RoBERTa), A Lite BERT (ALBERT).
Generative Models: Generative Pretrained Transformer (GPT-2, GPT-3), Text-to-Text Transfer Transformer (T5).
Fine-tuning and Transfer Learning.
HuggingFace Transformers Library.

Module – 4 Advanced Applications in NLP No. of hours 6

Question Answering (QA) Systems: Closed-domain QA and Open-domain QA.
Conversational AI: Dialogflow (by Google), Open-source Conversational Framework (Rasa). Summarization: Extractive and Abstractive.
Machine Translation: OpenNMT, MarianMT.
Real-World Case Studies: Healthcare, Finance, Legal NLP, etc.

Module – 5 Speech and Multimodal NLP No. of hours 6

Automatic Speech Recognition (ASR) using Wav2Vec 2.0 (by Facebook AI), Whisper (by OpenAI).
Text-to-Speech (TTS) with Tacotron 2 and FastSpeech 2.
Multimodal NLP: Contrastive Language-Image Pretraining (CLIP) – OpenAI, Flamingo (DeepMind), Large Language and Vision Assistant (LLaVA).
NLP for Low-Resource Languages Overview - Multilingual BERT (mBERT), Cross-lingual RoBERTa (XLM-R), IndicNLP.

Course outcomes

- a) Apply foundational and contextual NLP techniques using linguistic preprocessing, word embeddings, and classical models.
- b) Design and implement deep learning architectures for sequence modelling and text classification tasks.
- c) Analyze, fine-tune, and deploy transformer-based models for advanced NLP tasks: sentiment analysis, question answering, and summarization.

d) Develop and integrate real-world NLP applications including speech-based systems, machine translation, and dialogue systems.

Text Books

1.	Daniel Jurafsky & James H. Martin. Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models. Pearson, 3rd Edition (Draft), January 2025.
2.	Lewis Tunstall, Leandro von Werra, Thomas Wolf. Natural Language Processing with Transformers. O'Reilly Media. Revised Edition, July 2022, ISBN-10:1098136799, ISBN-13:978-1098136795.
3.	Philipp Koehn. Neural Machine Translation. Cambridge University Press, First Edition, November 2020, ISBN-10:108497322, ISBN-13:978-1108497329.
4.	Yoav Goldberg. Neural Network Methods for Natural Language Processing (Synthesis Lectures on Human Language Technologies). Morgan & Claypool Publishers, First Edition, 2017. ISBN-10: 1627052984, ISBN-13: <u>978-1627052986</u> .
5.	Daniel Bikel & Imed Zitouni. Multilingual Natural Language Processing Applications: From Theory to Practice. IBM Press, First Edition, May 2012, ISBN-10:0137151446, <u>ISBN-13:978-0137151448</u>
6.	Jacob Eisenstein. Introduction to Natural Language Processing (Adaptive Computation and Machine Learning series). The MIT Press, Illustrated Edition, October 2019, ISBN-10:0262042843, ISBN-13:978-0262042840.

Reference books

1.	Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana. Practical Natural Language Processing. O'Reilly Media, First Edition, July 2020, ISBN-10:1492054054, ISBN-13:978-1492054054.
2.	Delip Rao and Brian McMahan. Natural Language Processing with PyTorch. 'Reilly Media, First Edition, December 2019. ISBN-10: 1491978236, ISBN-13: 978-1491978238.
3.	Dipanjana Sarkar. Text Analytics with Python: A Practical Real-World Approach to Gaining Actionable Insights from Your Data. Apress, Second Edition, July 2019. ISBN-10: 1484243533, ISBN-13: 978-1484243531.
4.	Ian Goodfellow, Yoshua Bengio, Aaron Courville. Deep Learning (Adaptive Computation and Machine Learning series). The MIT Press. First Edition, November 2016, ISBN-10: 0262035618, ISBN-13: 978-0262035613.
5.	Steven Bird, Ewan Klein, and Edward Loper. Natural Language Processing with Python – Analyzing Text with the Natural Language Toolkit. O'Reilly Media, First Edition, June 2009. ISBN-10: 0596516495, ISBN-13: 978-0596516499.
6.	Christopher D. Manning and Hinrich Schütze. Foundations of Statistical Natural Language Processing. MIT Press, First Edition, June 1999. ISBN-10: 0262133601,

RV Vidyaniketan, 8th Mile, Mysuru Road, Bengaluru – 560059, India
Ph: +91 80 68199900 | www.rvu.edu.in

Lab Programs: Using Python and popular libraries like NLTK, spaCy, Hugging Face Transformers, and PyTorch/TensorFlow.	
1.	<i>Text Preprocessing:</i> Tokenization, stop-word removal, lemmatization, POS tagging, NER. Tools: spaCy, NLTK
2.	<i>Word Representation Techniques:</i> Compute BoW and TF-IDF vectors for a corpus. Compare with GloVe/Word2Vec embeddings. Tools: scikit-learn, genism.
3.	<i>Sequence Modelling for Text Classification:</i> Build a sentiment classifier using LSTM or GRU on IMDB or product reviews. Tools: Keras or PyTorch.
4.	<i>Attention and CNN for Text:</i> Implement attention mechanism or text classification using CNN. Tools: TensorFlow/Keras, PyTorch.
5.	<i>Fine-tuning BERT on Custom Dataset:</i> Text classification or NER using Hugging Face Transformers. Dataset: IMDb, AGNews, or any custom labelled dataset. Tools: Hugging Face Transformers, Datasets, scikit-learn.
6.	<i>Language Generation with GPT-2 or T5:</i> Generate text given a prompt or paraphrase sentences. Tools: Transformers, Pipeline, GPT2LMHeadModel.
7.	<i>Question Answering using Pretrained Models:</i> Use a SQuAD-style context-question-answer pipeline with BERT. Tools: Hugging Face, pipeline(task="question-answering").
8.	<i>Summarization:</i> Implement an abstractive text summarization system using state-of-the-art transformer models like BART or T5. Tools: Transformers, Hugging Face pipeline.
9.	<i>Machine Translation:</i> Translate sentences using MarianMT or mBART. Tools: Hugging Face Transformers Library.
10.	<i>Speech-to-Text using Whisper or Wav2Vec 2.0:</i> Transcribe speech into text from .wav files. Tools: OpenAI Whisper, Facebook Wav2Vec2, transformers.
11.	<i>Text-to-Speech with Tacotron or FastSpeech:</i> Convert text to speech and play audio output. Tools: TTS (Coqui), ESPNet, PyTorch, Tacotron.
12.	<i>Multimodal NLP Demo:</i> Use CLIP or LLaVA to generate captions or match images with text prompts. Tools: OpenCLIP, LLaVA, transformers, datasets.
13.	<i>Mini Project:</i> Project Proposal + Dataset Curation
14.	<i>Mini Project:</i> Model Training, Evaluation and Optimization
15.	<i>Mini Project:</i> Final Demo + Viva + Report Submission